



DR. ERUM MEHMOOD

Backend-focused Software Engineer & AI Systems Researcher

PROFESSIONAL SUMMARY

Backend-focused Software Engineer and AI Systems Researcher with 13+ years of experience in scalable backend development, distributed data systems, ETL pipelines, cybersecurity-oriented architectures, and intelligent automation workflows. Experienced in designing secure and high-performance applications using Python, Node.js, SQL/NoSQL databases, OCR pipelines, and real-time data processing frameworks. Proven ability to build and optimize APIs, automate complex document-processing systems, and develop scalable architectures for data-intensive applications. Strong background in AI-driven systems, distributed stream processing, and secure computing, with practical experience spanning software engineering, research, and technical mentorship.

PROFESSIONAL EXPERIENCE

Software Developer (Freelance) | Self-Employed (2014 – Present)

- Designed and developed scalable backend systems, RESTful APIs, and intelligent data-processing pipelines using Python, Node.js, and cloud-integrated workflows.
- Built OCR-based PDF parsing and ETL systems capable of extracting, validating, and transforming heterogeneous document formats into structured Excel and database outputs.
- Developed AI-driven automation solutions for large-scale data extraction, preprocessing, and workflow optimization.
- Implemented secure backend architectures, authentication workflows, and real-time processing mechanisms for web and data-centric applications.
- Worked with distributed processing concepts, streaming pipelines, and scalable storage mechanisms for handling large data volumes efficiently like AWS, PySpark, Apache Spark Streaming, Apache Kafka, and MongoDB.
- Optimized system performance by analyzing memory, CPU, and network bottlenecks at scale, improving overall processing efficiency and operational overhead.
- Collaborated with cross-functional teams and mentored junior developers and students on backend engineering, debugging, and software architecture practices.

Lecturer & Mentor | Various Academic Institutions (2017 – Present)

- Delivered comprehensive undergraduate and graduate level courses in Python, Java, C++, Artificial Intelligence, Data Structures, Big Data Analytics, and Software Engineering.
- Mentored students and development teams in building robust backend systems, APIs, mobile applications, AI-based solutions, and distributed computing projects.
- Supervised collaborative software projects involving OCR automation, cybersecurity simulations, ETL systems, and intelligent data-processing architectures.
- Guided students in systems design, debugging strategies, software development workflows, and scalable application architecture.
- Conducted active research and practical implementations in real-time

CONTACT

Email

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Location

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Portfolio

<https://erummehmood.github.io/portfolio-website/>

GitHub

<https://github.com/ErumMehmood>

LinkedIn

<www.linkedin.com/in/erum-mehmood>

Google Scholar

<https://scholar.google.com/citations?user=7TjoqbgAAAAJ&hl=en>

TECHNICAL SKILLS

• Backend Development:

Node.js, Express.js, Flask, FastAPI, REST APIs

• Programming Languages:

Python, JavaScript, Java, C++, C, C#, SQL

• Databases:

MongoDB, PostgreSQL, MySQL, SQLite

• AI & Data Engineering:

Pandas, TensorFlow, OCR, OpenCV, Apache Spark, PySpark, Apache Kafka, Big Data Analytics

• Security & Systems:

JWT Authentication, API Security, OCPP Security, Encryption, Wireshark, Intelligent Secure Systems

• DevOps & Tools:

AWS, Docker, Linux, Git/GitHub, CI/CD, Automated Workflows, Network Optimization

EDUCATION

PhD in Computer Science

University of Management and Technology (UMT), Lahore
2018 – 2024

Dissertation: Integrated Real-Time Distributed Stream-Disk Processing Architecture for Unstructured Big Data

MPhil in Computer Science

NCBA&E, Lahore
2014 – 2017

Research Focus: Stream Processing for Real-Time Data Warehousing

distributed stream-disk processing and intelligent secure systems.

FEATURED TECHNICAL PROJECTS

AI-Driven Multi-Format PDF Parsing & ETL System *(Python, OCR, Tesseract, PyMuPDF, Camelot, Pandas, OpenPyXL)*

- Developed an intelligent document-processing pipeline for extracting and transforming structured and semi-structured PDF data into Excel-based reporting systems.
- Implemented OCR workflows, table extraction, timestamp-aware ETL logic, and validation pipelines for heterogeneous document formats.
- Designed automated regression-validation workflows supporting high-accuracy data extraction.

EV Charging Security Simulation Platform *(Python, Docker, WebSockets, OCPP, Wireshark)*

- Developed a simulated EV charging security environment for analyzing cyber threats in OCPP-based charging ecosystems.
- Implemented attack simulation, traffic monitoring, and anomaly-detection workflows for cyber-physical systems research.

DNA-Based Secure Image Encryption System *(Python, OpenCV, NumPy)*

- Developed adaptive image-encryption mechanisms using hyper-chaotic systems, permutation-diffusion models, and nonce-based security techniques.
- Designed high-entropy encryption workflows resistant to statistical and keystream reuse attacks.

Healthy Empower Backend System *(Node.js, Express.js, SQLite, React Native)*

- Developed backend APIs, authentication modules, and reward-processing systems for a health-focused mobile application.
- Integrated database workflows and scalable API routing mechanisms.

RESEARCH & PUBLICATIONS

- Published peer-reviewed research in AI-driven cybersecurity, distributed stream processing, big data systems, and intelligent data architectures.
- Publications featured in reputable venues including IEEE Access and Springer journals.
- Active reviewer for IEEE Access and other international journals.